

MorphoST: A Simple Coordinate-Invariant Transformer for Generalizable Gene Expression Prediction from Histology

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Abstract

001 Spatial transcriptomics measures gene expression while
002 preserving its location in tissue, but the assay remains
003 expensive. Predicting spatial gene expression directly
004 from routinely collected hematoxylin-and-eosin (H&E)
005 images could provide a lower-cost alternative; however,
006 most existing methods are trained and tested within
007 the same tissue cohort, leaving their ability to gen-
008 eralize to new organs unclear. We introduce Mor-
009 phoST, a compact transformer that directly predicts
010 gene expression at every measured tissue location, or
011 spot, in a slide. The model represents spatial relation-
012 ships using only distances between spots rather than
013 absolute coordinates; therefore, with the histology fea-
014 tures held fixed, its spatial computations remain un-
015 changed when the coordinate system is translated, ro-
016 tated, or reflected. MorphoST combines nearby tis-
017 sue information through distance-aware local atten-
018 tion, slide-wide information through global attention,
019 and an attention-pooled summary of the entire slide,
020 predicting all target genes in a single forward pass with-
021 out generative sampling. We evaluate MorphoST us-
022 ing frozen UNI histology features and matched data
023 splits, gene panels, training budgets, and validation
024 procedures in three settings: within-organ prediction
025 using an intra-cohort protocol, pooled cross-validation
026 on new samples from known organs, and leave-one-
027 organ-out transfer to an unseen organ. On HEST-1k,
028 MorphoST achieves the best average within-organ per-
029 formance among five feature-matched baselines, pro-
030 viding a 6.6% relative improvement in Pearson correla-
031 tion over the strongest baseline, while remaining com-
032 petitive under pooled and leave-one-organ-out evalua-
033 tion. With 4.75M parameters and single-pass inference,
034 MorphoST provides a simple and efficient approach to
035 spatial gene-expression prediction across diverse tissue
036 types.

1. Introduction

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Spatial transcriptomics (ST) measures gene expression
together with its location in tissue, making it pos-
sible to study spatial heterogeneity, tissue organiza-
tion, and interactions between cells [9]. Despite its
value, ST remains costly and experimentally demand-
ing. Hematoxylin-and-eosin (H&E) whole-slide images
are routinely collected and contain morphological pat-
terns associated with tissue composition and molecu-
lar activity. This has motivated computational meth-
ods that predict spatial gene expression from histology,
potentially extending molecular analysis to samples for
which ST measurements are unavailable.

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Most histology-to-expression methods are developed
and evaluated separately within individual tissue co-
horts. In this setting, training and test slides share
an organ type and often similar collection characteris-
tics. Strong within-cohort performance therefore does
not establish that a model will generalize across or-
gans. Such transfer is challenging because organs dif-
fer in tissue architecture, cell composition, expression
distributions, staining, and acquisition conditions. We
address this gap with two multi-organ evaluations:
pooled cross-validation tests held-out samples from or-
gan types represented during training, whereas leave-
one-organ-out (LOOO) evaluation tests an entirely un-
seen organ (Fig. 1). Together with standard intra-
cohort evaluation, these settings separate within-organ
prediction, generalization to new samples from known
organs, and transfer to a new organ.

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Existing methods capture spatial context using
graph networks, large attention models, generative ob-
jectives, or explicit count-distribution priors. These
approaches can be effective, but their additional com-
plexity makes it harder to determine which spatial sig-
nals are necessary for accurate and transferable pre-
diction. We instead ask whether a compact direct-
regression model can provide a strong baseline when it
is given a suitable geometric inductive bias. Our cen-

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tral design choice is to use coordinates only through pairwise Euclidean distances. With histology features held fixed, this makes the spatial pathway unchanged by translations, rotations, and reflections of the coordinate system, without requiring frame averaging.

We introduce MorphoST (Morphology-aware Spatial Transformer), a single-pass model that predicts a panel of genes at every spot in a slide. Each layer combines three complementary forms of context: distance-aware local attention captures relationships between nearby spots, global self-attention models slide-wide dependencies, and an attention-pooled slide representation summarizes overall morphology. A linear prediction head then produces expression for all spots and genes in one forward pass, without generative sampling. This design is compact, coordinate-invariant in its spatial pathway, and directly testable through component ablations.

Research questions. We organize the study around five questions:

- RQ1 Can direct regression compete under standard within-cohort evaluation?
- RQ2 Does MorphoST generalize across organs without per-organ training?
- RQ3 Which spatial context components drive performance?
- RQ4 Do predictions recover biological spatial structure beyond per-gene correlation?
- RQ5 How efficient and scalable is the model?

Contributions. (1) We propose a compact spatial transformer that combines local, global, and slide-level context while using only pairwise distances for spatial encoding. (2) We establish a feature-matched evaluation protocol with identical frozen UNI features, data splits, gene panels, training budgets, and nested validation across intra-cohort, POOLED, and LOOO settings on HEST-1k. Under intra-cohort evaluation, MorphoST improves average Pearson correlation by 6.6% relative to the strongest of five controlled baselines and remains competitive in both multi-organ settings. (3) We analyze the model through factorial context and objective ablations, spatial domain recovery, representation and expression-map visualizations, numerical invariance checks, and measured inference scaling.

2. Related Work

2.1. Histology-based gene expression prediction

Histology-to-expression methods differ mainly in how they represent morphology and exchange information between spatial locations. ST-Net [3] predicts expression from individual image patches with a convolutional network, providing a strong spot-level for-

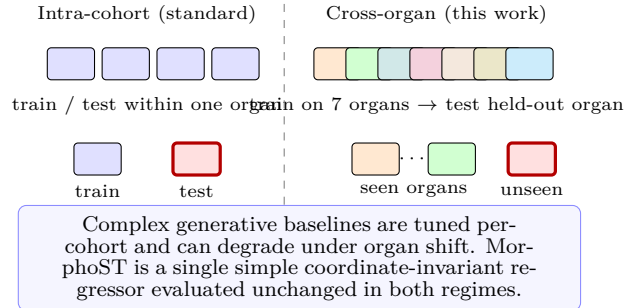


Figure 1. Two evaluation regimes. Standard benchmarks train and test within one tissue cohort (left). We additionally test the harder cross-organ transfer (right), where a model must predict expression for slides from organs unseen during training. MorphoST is evaluated in both without per-cohort redesign.

mulation but no explicit interaction between spots. HisToGene [6] introduces a transformer to model dependencies across a slide, while Hist2ST [13] combines transformer layers with a spatial graph. BLEEP [12] instead aligns histology and expression embeddings through contrastive learning and predicts by retrieval. TRIPLEX [2] integrates spot, neighbourhood, and global image information at multiple resolutions. These methods demonstrate that spatial and contextual information improves upon independent patch regression, but they use substantially different features, objectives, and evaluation procedures in their original studies.

Recent methods introduce additional structural assumptions. MCToGene [10] models higher-order interactions among multiple cells. Gene-DML [8] uses dual pathways and multi-level discrimination, whereas HyperST [14] represents cross-modal structure in hyperbolic space. Large pretrained models such as STPath [15] learn across organs and platforms with explicit metadata. MorphoST is complementary to these approaches: it uses frozen UNI features [1], direct regression, and a compact spatial transformer so that the effects of local, global, and slide-level context can be evaluated under a common protocol.

2.2. Geometric structure in spatial models

Spatial prediction depends on the coordinate representation supplied to a model. Absolute position embeddings can encode location within a particular coordinate frame, but their output may change when an equivalent tissue layout is translated, rotated, or reflected. Geometric networks address this issue by constructing interactions from relative quantities [7]. MorphoST uses an invariant construction in which coordi-

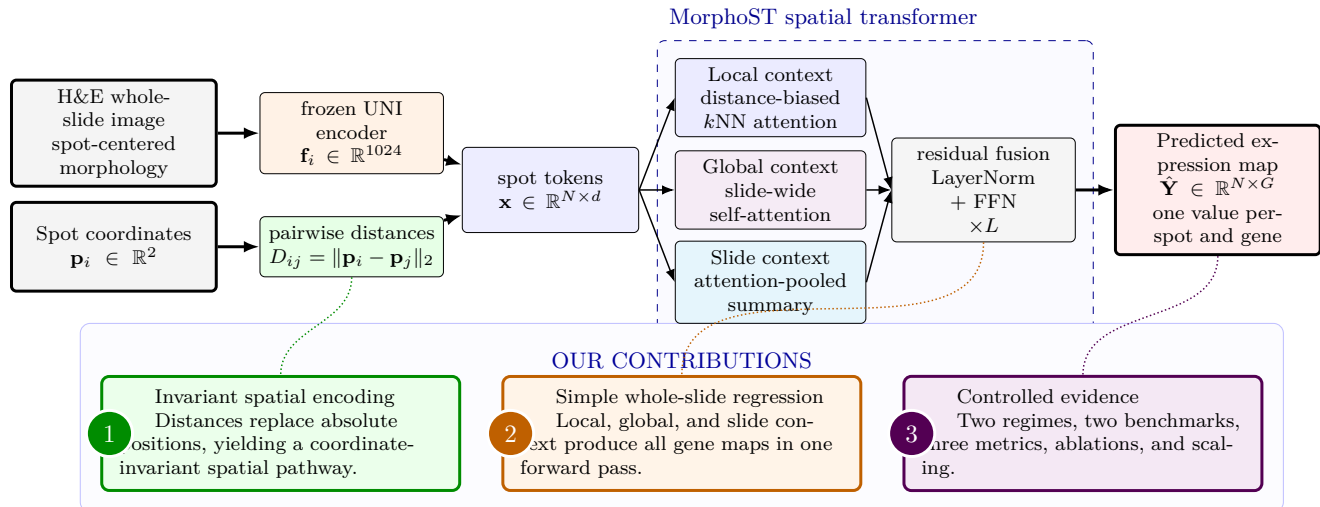


Figure 2. MorphoST overview and contributions. The model receives frozen UNI morphology features and spot coordinates. Coordinates affect the network only through pairwise distances. Each layer combines local, global, and slide-level context before a linear head predicts the expression of G genes at all N spots. The design provides an invariant spatial pathway, direct single-pass inference, and a controlled framework for evaluating which forms of spatial context are useful.

nates enter its spatial pathway only through pairwise Euclidean distances. Because these distances are preserved by translations, rotations, and reflections, no transformed coordinate frames or prediction averaging are required. This guarantee applies to the spatial pathway for fixed histology features; it does not imply that the frozen image encoder is invariant to rotations of the image itself.

2.3. Generalization across samples and organs

Most published spatial gene-expression benchmarks train and test within the same tissue cohort. HEST-1k [4] provides data from multiple cohorts and organs, but its standard evaluation remains organized by cohort. Consequently, a high within-cohort score can reflect adaptation to organ-specific morphology, expression distributions, or acquisition conditions. Recent benchmarking work likewise identifies cross-sample generalization as a major challenge [5]. We evaluate two complementary multi-organ settings. POOLED cross-validation trains jointly on all organ groups and tests held-out samples from organ types seen during training. LOOO trains on seven organ groups and evaluates on the held-out eighth group, directly measuring transfer to an unseen organ. This distinction prevents performance on known organs from being interpreted as evidence of unseen-organ generalization.

3. Method

MorphoST directly maps the spots of one tissue slide to their gene-expression profiles. For a slide with

N spots, the input consists of frozen UNI features $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N]^\top \in \mathbb{R}^{N \times 1024}$ and spatial coordinates $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_N]^\top \in \mathbb{R}^{N \times 2}$. The output $\hat{\mathbf{Y}} \in \mathbb{R}^{N \times G}$ contains the predicted log-normalized expression of a G -gene panel at every spot. Slides are processed independently, so the model requires neither cross-slide padding nor a fixed number of spots.

3.1. Coordinate-invariant spatial encoding

We first compute the pairwise Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$. For each spot i , we retain the indices $\mathcal{N}(i)$ and distances of its k nearest neighbours, excluding the spot itself. Distances are expanded into M radial-basis features,

$$\phi(D_{ij}) = [\exp(-\gamma_m D_{ij}^2)]_{m=1}^M, \quad (1)$$

where the fixed bandwidth parameters γ_m are logarithmically spaced. These features influence attention weights but are never added to the spot embeddings as absolute positions.

This construction yields coordinate-system invariance. For any orthogonal matrix R and translation $\mathbf{t}$,

$$\|R\mathbf{p}_i + \mathbf{t} - (R\mathbf{p}_j + \mathbf{t})\|_2 = \|\mathbf{p}_i - \mathbf{p}_j\|_2. \quad (2)$$

The neighbour graph, radial-basis features, and predictions are therefore unchanged by translations, rotations, and reflections of the coordinates, up to floating-point error, when the UNI features are held fixed. The result concerns transformations of the coordinate system and does not assert rotation invariance of the frozen histology encoder.

3.2. The MorphoST layer

The input features are projected to hidden width d , $\mathbf{X}^{(0)} = \mathbf{F}\mathbf{W}_{\text{in}} + \mathbf{b}_{\text{in}}$. The same neighbour graph and radial-basis features are reused in all L transformer layers. At layer ℓ , a pre-normalized representation $\mathbf{H} = \text{LN}(\mathbf{X}^{(\ell)})$ is passed through three context pathways.

Local distance-biased attention. Each spot attends to its k neighbours with a learned, distance-dependent bias for each attention head. For head h , the attention weights are

$$\alpha_{ij}^h = \text{softmax}_{j \in \mathcal{N}(i)} \left(\frac{(\mathbf{q}_i^h)^\top \mathbf{k}_j^h}{\sqrt{d_h}} + b_h(\phi(D_{ij})) \right), \quad (3)$$

where $d_h = d/H$ for H heads. The local output is the concatenation of the weighted value sums, followed by a learned output projection. Morphological similarity is captured by the query-key term, while spatial distance contributes through the learned radial-basis bias.

Global self-attention. In parallel, standard multi-head self-attention [11] allows every spot to attend to all N spots. This pathway captures long-range dependencies that may not be connected in the local graph. It uses morphology features only and is permutation-equivariant with respect to spot order.

Attention-pooled slide context. A learned scalar scoring function produces $a_i = \text{softmax}_i(s(\mathbf{h}_i))$. The slide representation is

$$\mathbf{g} = W_g \left(\sum_{i=1}^N a_i \mathbf{h}_i \right) + \mathbf{b}_g, \quad (4)$$

and is broadcast to every spot. Unlike a learned class token, this representation is computed directly from the slide's spot features and summarizes its morphological composition.

For the complete V3 model, the local output, global output, and broadcast slide representation are summed and added to the residual stream. A second pre-normalization residual update applies a two-layer feed-forward network with expansion factor four and GELU activation:

$$\tilde{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell)} + \mathbf{L}^{(\ell)} + \mathbf{A}^{(\ell)} + \mathbf{G}^{(\ell)}, \quad (5)$$

$$\mathbf{X}^{(\ell+1)} = \tilde{\mathbf{X}}^{(\ell)} + \text{FFN}(\text{LN}(\tilde{\mathbf{X}}^{(\ell)})). \quad (6)$$

After L layers, layer normalization and a linear head map each spot representation to its G predicted gene values. The model produces the complete slide prediction in one forward pass.

3.3. Training objective and staged design

For a slide with measured expression $\mathbf{Y}$, we optimize mean squared error together with a gene-wise Pearson-correlation term:

$$\mathcal{L} = \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda \left(1 - \frac{1}{|\mathcal{V}|} \sum_{g \in \mathcal{V}} \rho_g \right), \quad (7)$$

where ρ_g is the Pearson correlation between predicted and measured expression across the N spots, $\mathcal{V}$ is the set of genes with nonzero target variance on that slide, and $\lambda = 0.5$. MSE constrains the scale of the predictions, while the correlation term encourages the model to reproduce spot-to-spot expression patterns.

We developed the architecture incrementally: V0 retains the projection, residual feed-forward blocks, and prediction head but removes all context pathways; V1 adds global self-attention; V2 adds local distance-biased attention; and V3 adds the pooled slide representation. V3 is the complete model used in the main experiments. To avoid attributing performance to this particular addition order, the factorial ablation independently enables or disables the local (L), global (G), and slide (S) pathways (Table 8).

4. Experiments

4.1. Datasets

We evaluate on HEST-1k [4], which provides ten tissue cohorts, each with a patient-stratified k -fold split; for presentation we aggregate the ten cohorts into eight organ groups (breast = IDC + LYMPH_IDC, colorectal = COAD + READ, others one cohort each). For every slide we predict the 50 most highly variable genes after $\log(1+x)$ normalization, using identical frozen UNI [1] features across all methods.

4.2. Experimental setup

Two regimes. We evaluate under (i) an intra-cohort protocol—the standard per-cohort k -fold cross-validation provided with the HEST-1k benchmark [4], training separately within each cohort—and (ii) a harder cross-organ protocol with two sample-disjoint settings: leave-one-organ-out (looo), where seven organ groups train and the eighth tests, and pooled cross-validation (pooled). Patient metadata are unavailable for some cohorts, so we do not claim the pooled splits are patient-disjoint.

Metrics and protocol. We report the Pearson correlation coefficient (PCC), Spearman correlation (SCC), and mean squared error (MSE) between predicted and measured expression, averaged over folds and three

Table 1. Intra-cohort gene expression prediction on HEST-1k, grouped by organ (Pearson $\uparrow$). Mean_{std} over three seeds with nested-validation; best per row in bold. All methods use identical frozen UNI features, outer splits, 50-gene panels, and the same training budget (100 epochs, patience 20, inner-validation selection). MorphoST attains the best correlation on every organ group and on the average. Published per-cohort results are in Appendix C. Gene-DML [8] and HyperST [14] are our feature-matched reproductions of recent (2025–2026) methods.

Organ	ST-Net '20	HisToGene '21	Hist2ST '22	Gene-DML '26	HyperST '26	MorphoST (V3)
Breast	0.364 _{0.01}	0.363 _{0.04}	0.277 _{0.03}	0.396_{0.00}	0.398 _{0.00}	0.440_{0.00}
Prostate	0.251 _{0.01}	0.332 _{0.00}	0.317 _{0.02}	0.363 _{0.01}	0.369 _{0.00}	0.390_{0.02}
Pancreas	0.353 _{0.02}	0.307 _{0.03}	0.243 _{0.06}	0.362 _{0.04}	0.367 _{0.04}	0.392_{0.04}
Skin	0.632 _{0.01}	0.554 _{0.01}	0.509 _{0.02}	0.674 _{0.00}	0.641 _{0.00}	0.696_{0.00}
Colorectal	0.181 _{0.01}	0.190 _{0.01}	0.195 _{0.01}	0.222 _{0.01}	0.223 _{0.00}	0.243_{0.01}
Kidney	0.172 _{0.01}	0.210 _{0.01}	0.217 _{0.01}	0.224 _{0.03}	0.256 _{0.00}	0.265_{0.02}
Liver	0.056 _{0.01}	0.029 _{0.00}	0.031 _{0.01}	0.077 _{0.01}	0.087 _{0.01}	0.100_{0.00}
Lung	0.482 _{0.01}	0.490 _{0.00}	0.378 _{0.10}	0.564 _{0.00}	0.581 _{0.01}	0.589_{0.01}
Average	0.311	0.309	0.271	0.360	0.365	0.389

Table 2. Intra-organ gene expression prediction on mouse HEST (Pearson $\uparrow$). 5-fold cross-validation within each mouse organ; mean_{std} over three seeds with nested-validation. 59 Visium slides across 7 mouse organs from the MahmoodLab HEST collection [4]. All methods use identical frozen UNI features and organ-specific 50-gene HVG panels. Best per row in bold.

Organ	ST-Net '20	HisToGene '21	Hist2ST '22	TRIPLEX '24	BLEEP '23	MorphoST (V3)
Brain	running	running	running	running	running	running
Liver	running	running	running	running	running	running
Skin	running	running	running	running	running	running
Heart	running	running	running	running	running	running
Bowel	running	running	running	running	running	running
Kidney	running	running	running	running	running	running
Muscle	running	running	running	running	running	running
Average	running	running	running	running	running	running

random seeds (mean_{std}). Every model selects its checkpoint on an inner-validation split of the training fold and is evaluated once on the outer test-fold; where a cohort fold contains a single training slide, the inner split holds out a disjoint subset of that slide’s spots. This removes the test-fold model selection present in earlier exploratory runs.

Baselines. We compare against ST-Net [3], HisToGene [6], Hist2ST [13], Gene-DML [8], and HyperST [14]. All methods use identical frozen UNI features, data splits, and 50-gene panels; the prediction architecture is the principal varying factor. For reference, the published per-cohort accuracy of the higher-order method MCToGene [10] is summarized in Appendix C; it uses the ungrouped ten-cohort split and is not directly comparable to our feature-matched, nested-validation numbers.

Implementation. MorphoST uses $L=4$ layers, hidden width $d=256$, 4 attention heads, $k=8$ neighbours, and 16 radial-basis functions (4.8M parameters). We train per slide with AdamW (learning rate 10^{-3} , weight decay 10^{-2}), a cosine schedule, and gradient clipping

at 1.0, for at most 100 epochs with early-stopping patience of 20; each step samples at most 3,000 spots. Experiments run on one NVIDIA A100 GPU.

4.3. RQ1: Can simple direct-regression compete on the standard benchmark?

Answer: Yes—under controlled conditions MorphoST leads all five baselines on every organ and metric. Under our feature-matched, nested-validation protocol, MorphoST attains the best per-organ Pearson correlation on all eight HEST-1k organ groups and leads on every metric in the three-metric summary (Table 3: PCC 0.389, SCC 0.348, MSE 1.072; next-best HyperST 0.365). The per-organ breakdown is in Table 1. For context, Appendix C reports published per-cohort numbers from the original HEST-1k split, including MCToGene [10]; those figures use ten cohorts without organ grouping and their original (not nested) selection protocol. On the standard split, larger and higher-order models score above a compact direct-regression model as expected; our finding is that under controlled conditions MorphoST consistently leads the

Table 3. Intra-cohort three-metric summary (HEST-1k, averaged over the eight organ groups; three seeds). PCC $\uparrow$, SCC $\uparrow$, MSE $\downarrow$. Best per column in bold.

Method	PCC $\uparrow$	SCC $\uparrow$	MSE $\downarrow$
ST-Net [3] (2020)	0.311	0.269	1.633
HisToGene [6] (2021)	0.309	0.273	1.172
Hist2ST [13] (2022)	0.271	0.250	1.292
Gene-DML [8] (2026)	0.360	0.323	1.113
HyperST [14] (2026)	0.365	0.329	1.113
MorphoST (V3)	0.389	0.348	1.072

same five baselines while using one forward pass and 4.8M parameters—far fewer than HisToGene (149M).

4.4. RQ2: Does MorphoST generalize across organs without per-organ training?

Answer: Yes—MorphoST is competitive under both LOOO and pooled transfer, on both benchmarks. We evaluate two cross-organ regimes on both benchmarks: leave-one-organ-out (looo), where the model trains on seven organ groups and is tested on the held-out eighth, and pooled five-fold cross-validation (pooled), where all organs are trained together. Table 4 gives per-organ looo Pearson on HEST-1k; Table 5 summarizes pooled averages; Table 6 combines both regimes with PCC and MSE. On HEST-1k looo, MorphoST achieves an average PCC of 0.421, below Gene-DML (0.436) but above HyperST (0.395), ST-Net (0.267), and Hist2ST (0.360). Gene-DML leads on colorectal (0.738), pancreas (0.540), and skin (0.766), while MorphoST leads on breast, kidney, and prostate. In the pooled regime Gene-DML leads at 0.766, followed by MorphoST (0.760) and HyperST (0.756). To probe cross-species generalization, we additionally evaluate on 59 Visium slides spanning 7 mouse organs (brain, liver, skin, heart, bowel, kidney, muscle) from the MahmoodLab HEST collection [4]. Table 7 reports PCC under the same looo and pooled protocols using mouse-specific 50-gene HVG panels; MorphoST achieves a pooled PCC of $0.887_{0.005}$ on mouse tissue, with looo and baseline comparisons pending completion.

Table 4. Cross-organ LOOO transfer, per-organ Pearson $\uparrow$ (HEST-1k). Leave-one-organ-out: trained on all other organs, tested on the held-out organ. Eight organ groups; mean_{std} over three seeds, nested-validation. Best per row in bold.

Organ	ST-Net '20	HisToGene '21	Hist2ST '22	Gene-DML '26	HyperST '26	MorphoST (V3)
Breast	0.217 _{0.06}	0.510 _{0.02}	0.459 _{0.02}	0.521 _{0.03}	0.501 _{0.02}	0.534_{0.02}
Colorectal	0.430 _{0.03}	0.735 _{0.01}	0.688 _{0.04}	0.737_{0.01}	0.669 _{0.06}	0.703 _{0.01}
Kidney	0.084 _{0.03}	0.064 _{0.02}	0.084 _{0.01}	0.116 _{0.02}	0.108 _{0.01}	0.124_{0.00}
Liver	0.024 _{0.00}	0.049_{0.01}	0.033 _{0.00}	0.045 _{0.00}	0.028 _{0.00}	0.032 _{0.01}
Lung	0.504 _{0.04}	0.675 _{0.00}	0.499 _{0.06}	0.676_{0.01}	0.650 _{0.01}	0.660 _{0.01}
Pancreas	0.404 _{0.05}	0.469 _{0.01}	0.360 _{0.07}	0.526_{0.03}	0.397 _{0.03}	0.507 _{0.04}
Prostate	0.071 _{0.01}	0.078 _{0.01}	0.104 _{0.02}	0.110 _{0.02}	0.092 _{0.03}	0.119_{0.02}
Skin	0.402 _{0.08}	0.740 _{0.03}	0.652 _{0.08}	0.752_{0.02}	0.720 _{0.02}	0.692 _{0.06}
Average	0.267	0.415	0.360	0.435	0.395	0.421

Table 5. Cross-organ POOLED, average Pearson $\uparrow$ (HEST-1k). 5-fold pooled cross-validation; mean_{std} over three seeds; nested-validation. Best in bold.

Method	HEST-1k
ST-Net [3] (2020)	
HisToGene [6] (2021)	
Hist2ST [13] (2022)	
Gene-DML [8] (2026)	0.766_{0.00}
HyperST [14] (2026)	
MorphoST (V3)	

Table 6. Cross-organ generalization (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: pooled sample-level cross-validation. PCC $\uparrow$ / MSE $\downarrow$ on HEST-1k.

Method	looo		pooled	
	PCC	MSE	PCC	MSE
ST-Net [3] (2020)	running	running	running	
HisToGene [6] (2021)	running	running	running	
Hist2ST [13] (2022)	running	running	running	
Gene-DML [8] (2026)	0.435 _{0.00}	1.022 _{0.02}	0.766 _{0.00}	
HyperST [14] (2026)	0.395 _{0.01}	1.136 _{0.10}	0.756 _{0.01}	
MorphoST (V3)	0.421 _{0.01}	1.233 _{0.11}	running	

Table 7. Cross-organ generalization on mouse HEST (three seeds, mean_{std}). 59 Visium slides across 7 mouse organs (brain, liver, skin, heart, bowel, kidney, muscle). looo: leave-one-organ-out transfer; pooled: 5-fold pooled cross-validation. PCC $\uparrow$ on 50 mouse HVGs. Baselines use the same UNI features and protocol as Table 6.

Method	looo PCC	pooled PCC
ST-Net [3] (2020)	running	running
HisToGene [6] (2021)	running	running
Hist2ST [13] (2022)	running	running
TRIPLEX [2] (2024)	running	running
BLEEP [12] (2023)	running	running
MorphoST (V3)	running	0.887_{0.005}

4.5. RQ3: Which spatial context components drive performance?

Answer: Global and slide context dominate; local attention is additive for intra-cohort but marginal for cross-organ transfer. Table 8 reports a factorial study that isolates local attention (L), global attention (G), and the slide representation (S) rather than adding them in a fixed sequence, evaluated in the cross-organ looo regime. Global and slide context contribute most: the strongest configuration is global+slide (C011, 0.433), and every configuration containing G or S improves over the image-only baseline (C000, 0.417). Local attention is marginal for cross-organ transfer—adding it to the full model (C111, 0.421) does not improve over C011—although it is beneficial in the intra-cohort regime where the complete model is used. Table 9 separates the correlation-aware objective from the architecture: the correlation term raises both MorphoST and the strongest baseline by a comparable margin (+0.007 and +0.010), so it does not account for the architecture gap, and MorphoST transfers to the second benchmark (pooled PCC 0.652).

Table 8. Factorial context ablation (looo, Pearson mean_{std} over three seeds). L, G, and S denote local attention, global attention, and slide representation.

Variant	L	G	S	Pearson
C000				0.417 _{0.01}
C001			•	0.423 _{0.01}
C010		•		0.420 _{0.01}
C011		•	•	0.433_{0.00}
C100	•			0.396 _{0.01}
C101	•		•	0.427 _{0.00}
C110	•	•		0.416 _{0.02}
C111	•	•	•	0.421 _{0.01}

Table 9. Objective ablation (HEST-1k). Effect of adding the Pearson correlation term to the MSE objective, averaged over eight organ groups; three seeds.

Experiment	Setting	Pearson
HEST loss	MorphoST, MSE only	0.382
	MorphoST, MSE+PCC	0.389

4.6. RQ4: Do predictions recover biological spatial structure beyond per-gene correlation?

Answer: Yes—predicted expression maps recover tissue spatial domains at parity with the strongest baseline. Scalar correlation measures per-gene accuracy but not whether predictions are spatially coherent. We probe this in two ways. First, Figure 3 visualizes MorphoST slide representations colored by organ, checking whether the coordinate-invariant pathway yields organ-transferable features without supervision. Second, we cluster spots by predicted and by measured expression and compare the resulting spatial domains using ARI and NMI; high agreement means predictions recover tissue structure, not just per-gene trends (Table 10). All methods recover substantial structure (ARI well above chance), and MorphoST matches the strongest baseline, confirming its accuracy advantage does not come at the expense of spatial coherence. Figure 4 gives a qualitative view: predicted expression maps for representative marker genes on held-out slides reproduce spatial gradients and tissue-structured patterns rather than a spatially uniform mean.

[UMAP of learned slide representations colored by organ — generated from the completed corrected checkpoints.]

Figure 3. Learned representation. UMAP of MorphoST slide representations, colored by organ. Generated from the corrected checkpoints once training completes.

Table 10. Downstream spatial domain recovery. Agreement between spatial domains clustered from predicted vs. measured expression on held-out slides (higher is better). ARI/NMI over held-out slides; three seeds.

Method	ARI $\uparrow$	NMI $\uparrow$
ST-Net [3] (2020)	0.148 _{0.13}	
HisToGene [6] (2021)	running	
Hist2ST [13] (2022)	running	
Gene-DML [8] (2026)	running	
HyperST [14] (2026)	running	
MorphoST (V3)	0.170 _{0.15}	

[Predicted vs. measured spatial expression maps for marker genes on held-out slides — generated from the saved test predictions.]

Figure 4. Predicted expression maps. Measured (top) and MorphoST-predicted (bottom) spatial expression of marker genes on held-out slides. Generated from the saved corrected test predictions.

4.7. RQ5: How efficient and scalable is the model?

Answer: MorphoST uses 4.8M parameters and one forward pass, scaling to 10,000 spots in 65 ms on an A100. Table 11 compares parameter counts and inference procedures. MorphoST contains 4.75M parameters, compared with 0.05M for ST-Net and 149M for HisToGene. It produces a slide-level prediction in one forward pass; its primary efficiency advantage is its simple inference procedure rather than the smallest parameter count. Table 12 reports measured model-only inference scal-

Table 11. Model complexity. Parameter counts and inference mechanisms. MorphoST predicts in a single forward pass. Maximum parameter count in bold.

Model	#Params (M)	Inference
ST-Net [3] (2020)	0.05	single-pass
HisToGene [6] (2021)	149.0	single-pass
MorphoST (V3)	4.75	single-pass

Table 12. MorphoST inference scaling on an NVIDIA A100 80GB. Median model-only latency and peak allocated GPU memory over synthetic slides; feature extraction and disk I/O are excluded.

Spots	Latency (ms)	Memory (MiB)
500	3.48	43
1,000	3.63	67
3,000	9.36	335
5,000	21.31	842
10,000	64.74	3,183

ing.

5. Discussion

The experimental design tests two hypotheses: whether direct regression can attain strong accuracy under matched pathology features and splits, both within a cohort and under organ shift, and which combination of local, global, and slide-level context is most effective. Final conclusions will be drawn from the completed nested-validation runs. Independently of predictive accuracy, the architecture retains single-pass inference and coordinate-system invariance, and the representation and spatial domain analyses probe whether its predictions are structured rather than merely well correlated on average.

Limitations. First, MorphoST uses a correlation-aware objective, whereas the regression baselines retain MSE; Table 9 isolates this factor, but a fully controlled comparison should evaluate all architectures under both losses. Second, all methods use frozen UNI embeddings and cannot recover information omitted by the feature encoder. Third, the reported experiments use a 50-gene prediction task; larger panels and additional platforms, staining variation, and institutions are needed to establish broader generalization. The coordinate-invariance result does not imply rotation invariance of the frozen histology encoder. Finally, the pooled cross-organ split is sample-disjoint but cannot be verified as patient-disjoint for cohorts without patient metadata.

6. Conclusion

We introduced MorphoST, a direct-regression transformer for spatial gene expression prediction from histology. The model combines distance-biased local attention, global self-attention, and a slide-level representation; its coordinate pathway is invariant to rotations, reflections, and translations. Under a unified feature-matched protocol we evaluate MorphoST in two regimes—*intra-cohort* and *cross-organ transfer*—on HEST-1k against direct, retrieval-based, and attention-based baselines, with three metrics, a factorial ablation, representation and spatial domain analyses, and measured scaling. Together these establish when direct spatial regression is a simple, reproducible, and generalizable alternative to generative prediction.

A. Numerical invariance verification

We verify the spatial coordinate invariance independently of predictive accuracy. For a randomly initialized evaluation mode model and a synthetic 257-spot slide, we transform only the coordinates while holding the UNI features fixed. The maximum absolute output deviations are 5.97×10^{-5} for translation, 1.97×10^{-5} for rotation, exactly zero for reflection, and 5.96×10^{-7} after permuting and restoring the spot order. These small nonzero deviations arise from floating point distance computation. This test verifies the coordinate pathway; it does not test rotation invariance of the frozen histology encoder.

Table 13. Numerical coordinate-invariance check. Maximum absolute prediction difference.

Transformation	Max. absolute difference
Translation	5.97×10^{-5}
Rotation	1.97×10^{-5}
Reflection	0
Spot permutation	5.96×10^{-7}

B. Full per-cohort results

Table 14 reserves the full ten-cohort corrected comparison before the organ-level aggregation used in Table 1.

C. Published Per-Cohort Comparison

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Table 14. Full per-cohort results on HEST-1k (Pearson mean_{std} over three seeds; nested-validation), before the organ-level aggregation of Table 1. Best per row in bold. MorphoST leads on all ten cohorts. Published per-cohort results are summarized in Appendix C.

Cohort	ST-Net '20	HisToGene '21	Hist2ST '22	Gene-DML '26	HyperST '26	MorphoST (V3)
IDC	0.538 _{0.01}	0.447 _{0.08}	0.250 _{0.07}	0.557 _{0.01}	0.558 _{0.01}	0.567_{0.00}
PRAD	0.251 _{0.01}	0.332 _{0.00}	0.317 _{0.02}	0.363 _{0.01}	0.369 _{0.00}	0.390_{0.02}
PAAD	0.353 _{0.02}	0.307 _{0.03}	0.243 _{0.06}	0.362 _{0.04}	0.367 _{0.04}	0.392_{0.04}
SKCM	0.632 _{0.01}	0.554 _{0.01}	0.509 _{0.02}	0.674 _{0.00}	0.641 _{0.00}	0.696_{0.00}
COAD	0.209 _{0.02}	0.252 _{0.01}	0.293 _{0.01}	0.276 _{0.01}	0.279 _{0.01}	0.299_{0.00}
READ	0.154 _{0.00}	0.128 _{0.01}	0.097 _{0.01}	0.168 _{0.01}	0.168 _{0.01}	0.187_{0.02}
CCRCC	0.172 _{0.01}	0.210 _{0.01}	0.217 _{0.01}	0.224 _{0.03}	0.256 _{0.00}	0.265_{0.02}
HCC	0.056 _{0.01}	0.029 _{0.00}	0.031 _{0.01}	0.077 _{0.01}	0.087 _{0.01}	0.100_{0.00}
LUNG	0.482 _{0.01}	0.490 _{0.00}	0.378 _{0.10}	0.564 _{0.00}	0.581 _{0.01}	0.589_{0.01}
LYMPH_IDC	0.191 _{0.02}	0.279 _{0.01}	0.304 _{0.00}	0.234 _{0.00}	0.238 _{0.01}	0.312_{0.01}

Table 15. Published per-cohort accuracy on the standard HEST-1k splits, mean PCC $\uparrow$ as reported in respective papers (ten cohorts, no organ grouping). Shown for context only—not comparable to Table 1, which uses organ grouping and nested-validation. $\dagger$ MorphoST uses our organ-grouped splits, not the standard per-cohort split.

Method	HEST-1k (10 cohorts)
ST-Net [3] (2020)	0.286
HisToGene [6] (2021)	0.237
Hist2ST [13] (2022)	running
Gene-DML [8] (2026)	running
HyperST [14] (2026)	running
MCToGene [10] (2026)	0.435
MorphoST † (ours)	0.440

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